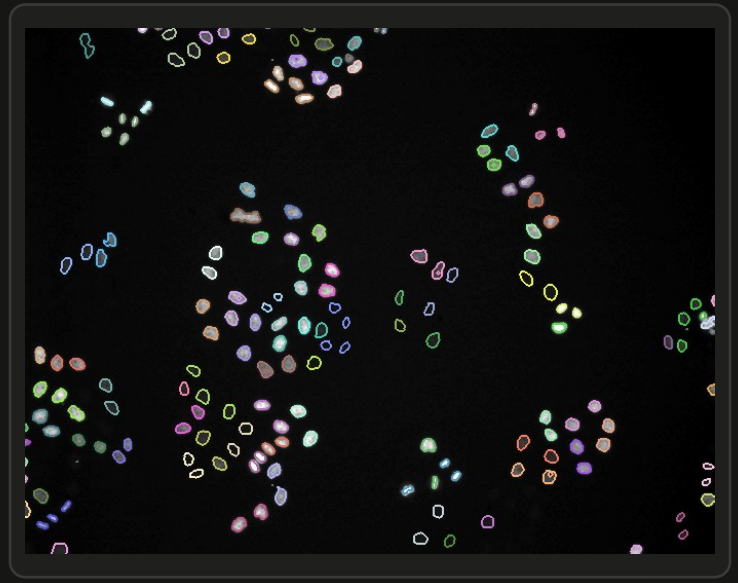
Can an AI agent run, and rerun, a bioimage analysis?

One 2014 ilastik workflow, five Claude sessions

THE BLIND RUNS' WHOLE BRIEF

“I want to answer group 5 task question, use ilastik and the mitocheck dataset”

+ the assignment PDF (the group question: “Can consensus become ground truth?”). No repository, no notes.



Frame 40 of the MitoCheck movie: each lineage in its own colour

What an agent session is

It gets a computer

In a chat, Claude only writes text. Here it can also run commands, install software, and read and write files.

It loops: command → result → next command

It reads error messages and tries again, on its own, until the task is done or it gets stuck.

The person watches and can step in

They answer questions and can step in at any time.

5

agent sessions

October 2026 · five sessions · same data

The data and the workflow

53-frame movie

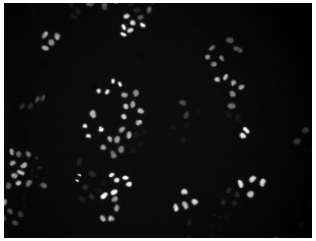
of dividing cell nuclei: ilastik's MitoCheck example

2014 project 1

pixel classification, trained by hand: finds nuclei

2014 project 2

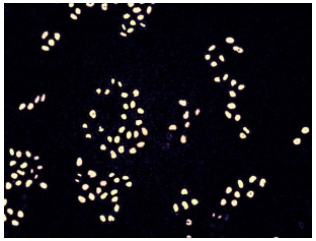
tracking, trained by hand: follows them



raw frame

the movie, frame 26

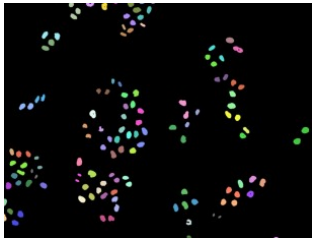
→



probability map

pixel classification

→

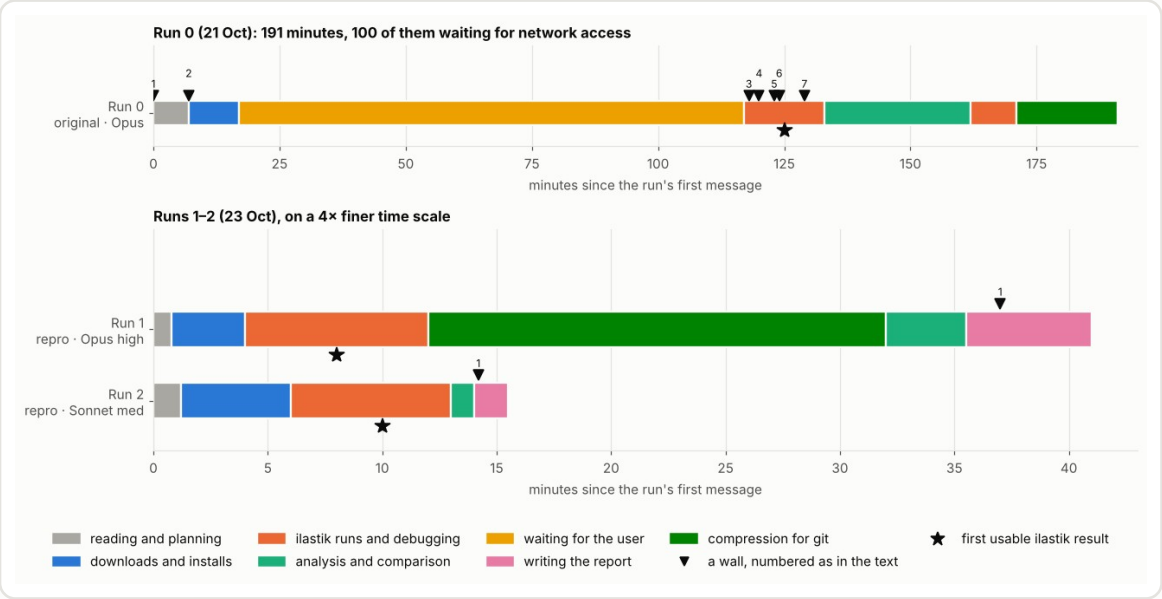


tracking labels

tracking

Lineage: a nucleus and all its descendants. Each lineage gets its own colour; grey objects were detected but not tracked. Run headless: from the command line, with no window.

Runs 0–2: the same workflow, three times



Colours are activities. Runs 1–2 are drawn on a 4× finer time scale.

191 min

Run 0: 100 of them waiting for network access; 7 walls

0 of 7

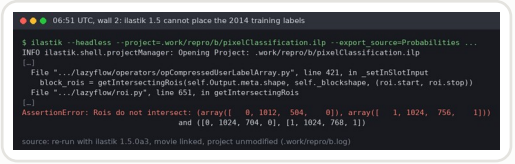
walls met by Runs 1–2, in 41 and ≈15 min, with no help

▼ **a wall**

★ **first usable ilastik result**

Run 0: the seven walls and their fixes

WALL	FIX
Network to ilastik's website blocked	→ **The person allowed the site**
ilastik 1.4.1 no longer installs with the usual package manager	→ **Newer pre-release, 1.5.0a3**
1.5.0a3 reads the 2014 labels with rows and columns swapped	→ **Small fix script**
Tracking settings kept where current versions no longer look	→ **Small fix script**
1.3.1 (the person's suggestion) is not in the public archive	→ **Tried 1.3.2post1**
1.3.2post1 crashed: the computer's system libraries, not the project	→ **Settled on 1.3.3post3**

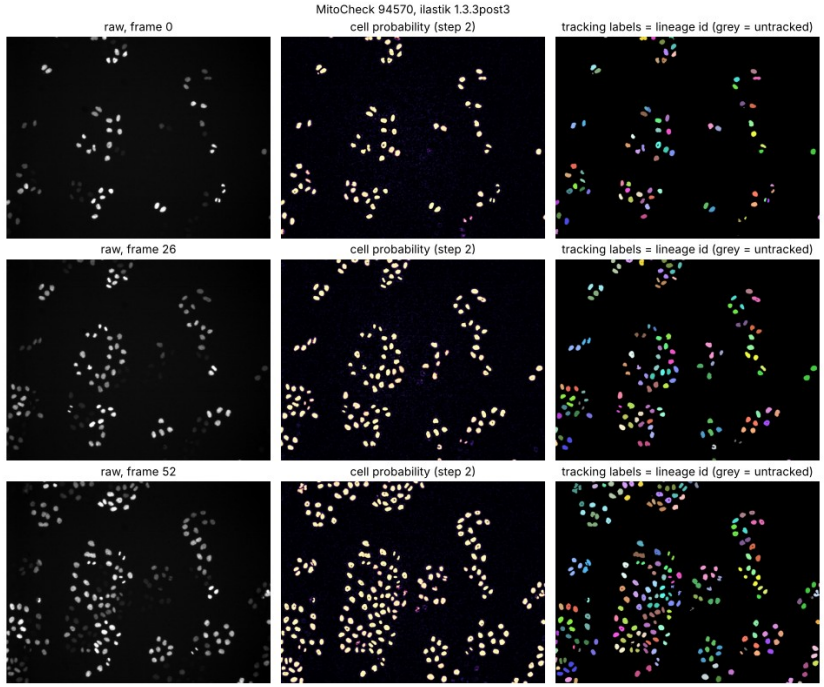


The labels wall: 1.5.0a3 cannot place the 2014 training labels

Plus two unchecked claims

That the versions numbered lineages differently; that 1.3.2post1 "crashes on this project". A second look caught both, and the text was corrected.

Run 0: the result



7,485
detections over the 53 frames

357
tracks

117
lineages

120
divisions

Both versions give the same counts. Their label images differ in 161 of 3.6 million pixels.

Runs 1 and 2 (23 Oct): reproduce it with the notes

Two fresh sessions were pointed at Run 0's repository and told to regenerate the results, not copy them.

Walls: who hit what

	Session 0 original · Opus	Session 1 repro · Opus high	Session 2 repro · Sonnet med	Session 3 blind · Sonnet med	Session 4 blind · Opus high
network to ilastik.org blocked	hit	–	–	–	–
conda: ilastik 1.4.1 won't install	hit	fixed in repo scripts	fixed in repo scripts	–	–
raw-data path relative to the .ilp	hit	fixed in repo scripts	fixed in repo scripts	–	avoided own way
headless opens projects read-only	–	–	–	–	hit
label axis order t,x,y vs t,y,x	hit	fixed in repo scripts	fixed in repo scripts	–	hit
tracking needs the 2014 probability map	hit	fixed in repo scripts	fixed in repo scripts	–	–
tracking parameters one level deeper (size_range)	hit	fixed in repo scripts	fixed in repo scripts	–	–
1.3.2post1 crashes on current glibc	hit	fixed in repo scripts	fixed in repo scripts	–	–
git push refused (repo not attached)	–	hit	hit	–	–
Python packages missing	–	–	–	hit	hit
tool limits (timeout, memory, own shell)	–	–	–	–	hit
wrong belief: "ilastik is GUI-only"	–	–	–	hit	–

Orange: hit · Blue: fixed by Run 0's scripts · Green: avoided · Grey: not met

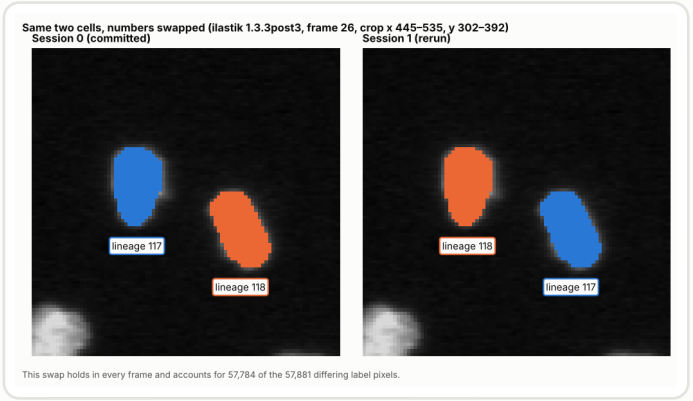
0 — times the person was needed

0 of 7 — Run 0's walls met

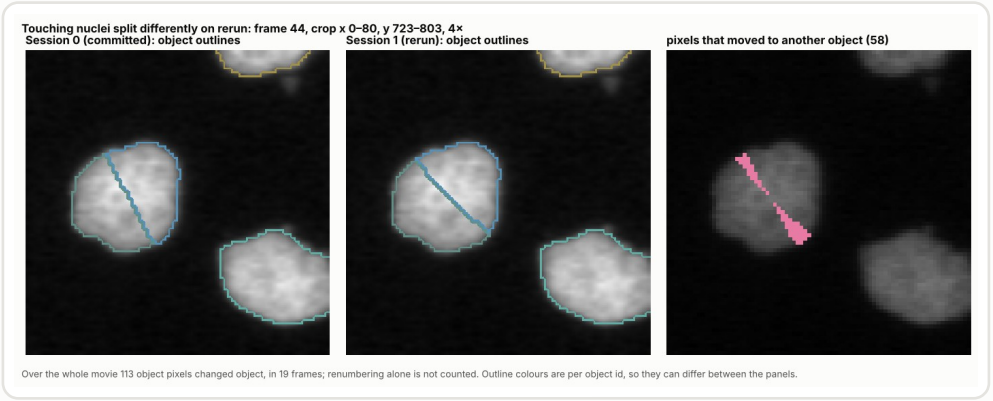
≈15 min — Sonnet, medium effort

41 min — Opus, high effort: mostly compressing files

Reproducible, but one step not exactly repeatable



Two lineages swap numbers (Run 0 vs Opus rerun)



The border between two touching nuclei moves

identical

1.5.0a3, both reruns: tracking labels pixel for pixel

113 px

1.3.3post3, Opus rerun: two lineages renumbered; 80 of the pixels at the touching pair

50 px

1.3.3post3, Sonnet rerun: pixels that differed

Runs 3 and 4: the same question, without the notes

WHAT THEY GOT

The assignment PDF, with the group's question:

"Can consensus become ground truth?"

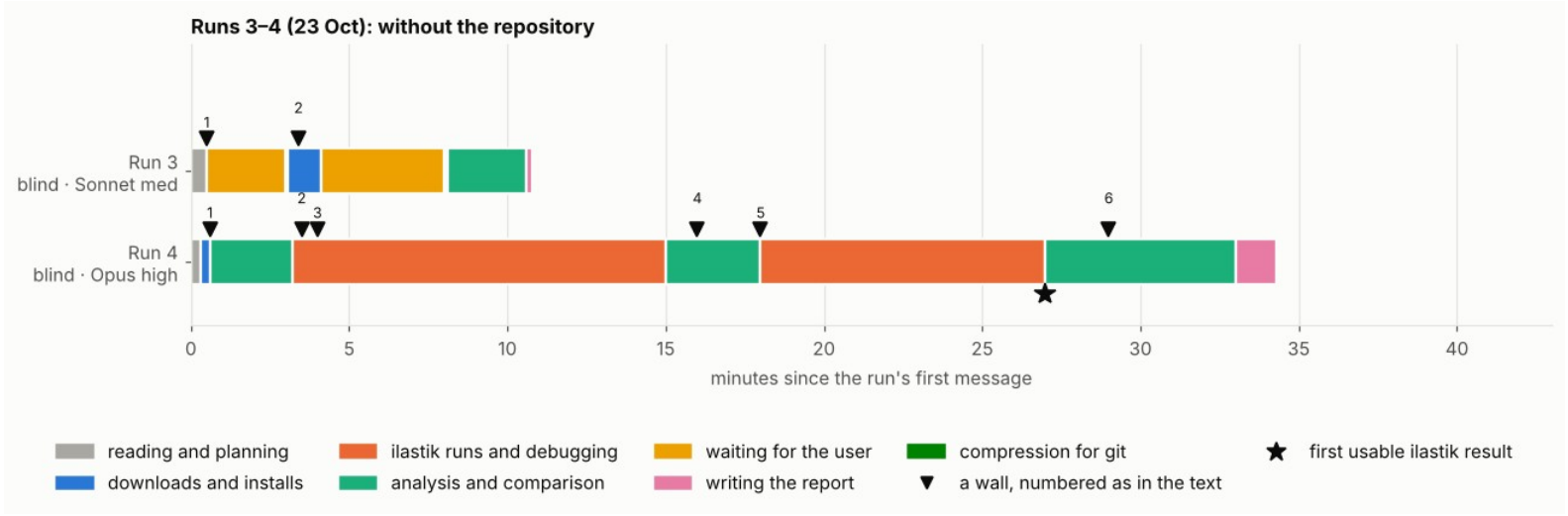
and one line: use ilastik and the MitoCheck dataset.

They could not see the repository

Neither ran the example's tracking; each built a consensus experiment

Their strategies parted in the first minute

Runs 3–4: the question, the data, nothing else



Run 3 (Sonnet) — 11 min, 6 of them waiting for the person; no ilastik

Run 4 (Opus) — 34 min, no questions; ilastik 1.4.2, headless

★ = first retrained classifier. Neither ran the example's tracking.

How ilastik was used, run by run

RUN	RAN ILASTIK?	PIXEL CLASSIFICATION	TRACKING
Runs 0, 1, 2	**yes**: 1.3.3post3 and 1.5.0a3	the 2014 project, as is	the 2014 project, as is
Run 4 (Opus)	**yes**: 1.4.2	the 2014 project retrained 8× with new strokes (--retrain)	**no**
Run 3 (Sonnet)	**no**	re-implemented outside ilastik (scikit-learn)	**no**



“Use ilastik” fixed the tool, not the workflow: the runs with the repository ran the example; Run 4 used ilastik as a trainable classifier; Run 3 kept only ilastik's files.

The commands

```
ilastik --headless --project=pixelClassification.ilp --export_source=Probabilities
```

→ Runs 0–2: open the 2014 project and write a probability map; nothing retrained

```
ilastik --headless --project=conservationTracking.ilp --raw_data=… --prediction_maps=…
--export_source=Tracking-Result
```

→ Runs 0–2: run the 2014 tracker on that map; nothing retrained

```
ilastik --headless --readonly=False --retrain --project=A1_tight.ilp
--export_source=Probabilities
```

→ Run 4: a copy of the 2014 project with the strokes replaced and the old classifier deleted; ilastik trains a fresh classifier and exports probabilities (8 times, ~1–1.5 min each)

From the session scripts. All three run ilastik headless: from the command line, with no window.

Run 3 (Sonnet): 11 minutes, asked twice

1. Read the PDF and found the group's question.
2. **No ilastik installed, so it concluded ilastik needs a window and cannot run here. It never checked.**
3. Guessed the data was on IDR (saying so), asked; the person pointed to ilastik's website.
4. Found no multiple annotators, asked; told to use the existing labels.
5. Rebuilt the classifier in scikit-learn; five annotators, each trained on a random 16 of the 27 label blocks.
6. Compared five rules on seven frames.

0.87–0.88

Fleiss' kappa (1 = perfect agreement)

1.9% → 4.8%

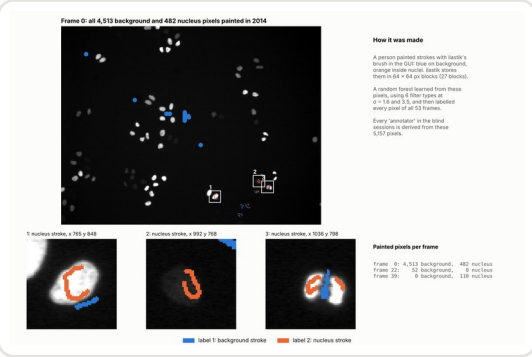
disputed pixels, frame 0 to frame 52

Consensus "is necessary for measuring reliability, but not sufficient to define ground truth"

Run 3's conclusion

Run 4 (Opus): 34 minutes, no questions

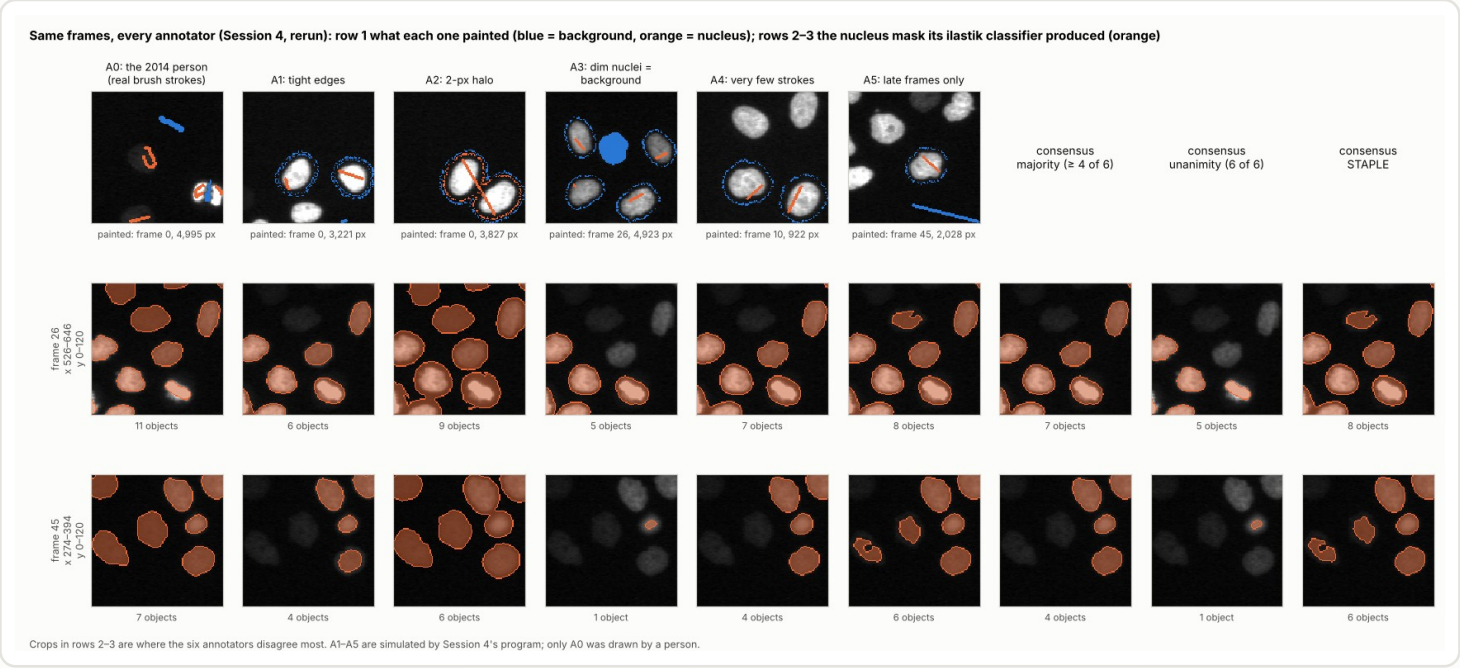
1. **Went straight to the download page; within a minute it was fetching the data and ilastik 1.4.2. Headless from then on.**
2. Opened the 2014 project: label 1 background, label 2 nucleus; 5,157 painted pixels.
3. Kept the real labels as A0; a program invented five more (A1–A5), each retrained headless.
4. Four walls, all fixed alone, among them rows and columns swapped: Run 0's trap.
5. Compared seven rules over the whole movie, STAPLE among them.
6. Added a biased pool; checked if rejected nuclei persist in neighbouring frames.



The 2014 brush strokes: 5,157 pixels, almost all on frame 0 (blue background, orange nucleus)

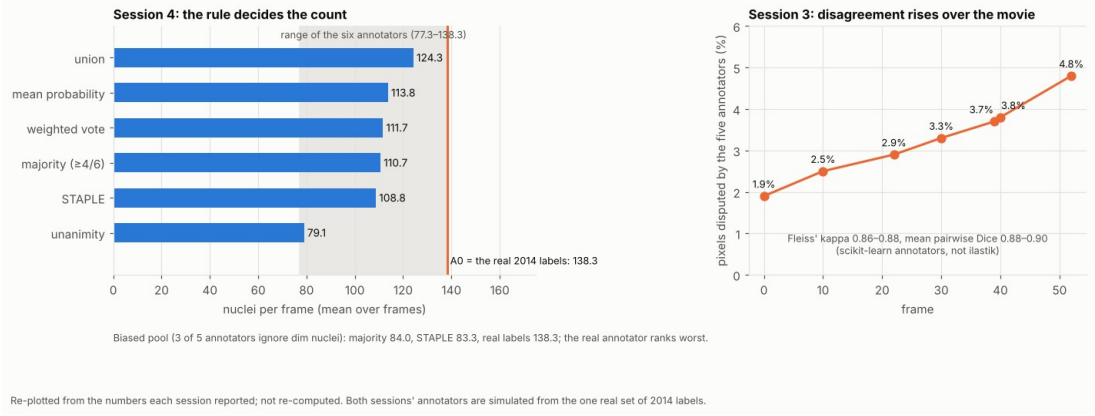
Invented: A1 tight edges · A2 two-pixel halo · A3 dim nuclei are background · A4 very few strokes · A5 late frames only

Six annotators, same frames



Row 1: what each annotator painted. Rows 2–3: frames 26 and 45, each nucleus mask and three consensus rules. A1, A3 and A4 miss most of the dim nuclei that A0 finds; majority vote drops them.

The rule decides the count



STAPLE: estimates how reliable each annotator is from their agreement alone, then trusts the reliable ones more.

NUCLEI PER FRAME

A0, real labels	**138**
simulated	**77–123**
union	**124**
majority	**111**
STAPLE	**109**
unanimity	**79**

84 · 83 vs 138

biased pool: majority, STAPLE vs A0

3,065 of 7,036

A0 objects rejected by the majority

94%

persist in both neighbouring frames

What a second look found

Run 3

- Annotators were one person's labels resampled, so their agreement is optimistic
- Its chronicle lists a step the transcript does not show

Run 4

- Annotators painted on objects found by a brightness cut-off, so the classifiers partly learn it back
- Treated the 2014 labels as correct without showing it
- Never asked whether real annotations existed

Rerun of Run 4

Within about 1%, except two:

108 vs 99
A4, nuclei per frame

111 vs 102
majority, nuclei per frame

Likely reason: A4 was trained on 922 pixels, and with a 4-of-6 threshold that one annotator moves the majority.

Two strategies, side by side

	Run 3 · Sonnet, medium effort	Run 4 · Opus, high effort
Where it looked first	Inward, at the computer: found an absence and never revisited it	Outward, at the download page: found a tool
Asking	Asked twice; lost six of its eleven minutes waiting	Asked nothing: saved time, but left the real question unasked
Effort	About $0.65 and 11 minutes	About $2.62 and 34 minutes; a stronger design
Same answer, different depth	Voting rules shift the result	Voting can reject the annotator most likely right

Both concluded that consensus is not sufficient. Neither used independent human annotators, which is what the question is really about.

What the five runs show

1. **Documentation turns a problem into a recipe.** With Run 0's notes, the reproductions met none of its seven walls; neither needed help.
2. **The first judgement decides the rest.** Run 4 looked at the download page; Run 3 looked at the computer and built on "ilastik cannot run here".
3. **Agents choose the experiment when the task is open.** Neither blind run ran the example's tracking; both designed a consensus experiment.
4. **Every session overstated something.** A second look, by a person or by another session, caught each one.
5. **Numbers can mislead.** 99.99% of pixels match the 2014 map, but an empty image scores 95%; nuclei overlap is 99.8%, and that is not ground truth either.

https://globias-2026-group5.github.io/

Runs 0–2 kept every command, error and output; Runs 3 and 4 committed nothing.

Analyses run by Claude (Anthropic) agent sessions; data courtesy of the MitoCheck project; example and projects from ilastik.